

Control Systems (EEE/ECE/INSTR F242)

Tutorial 12 (21 April 2026)

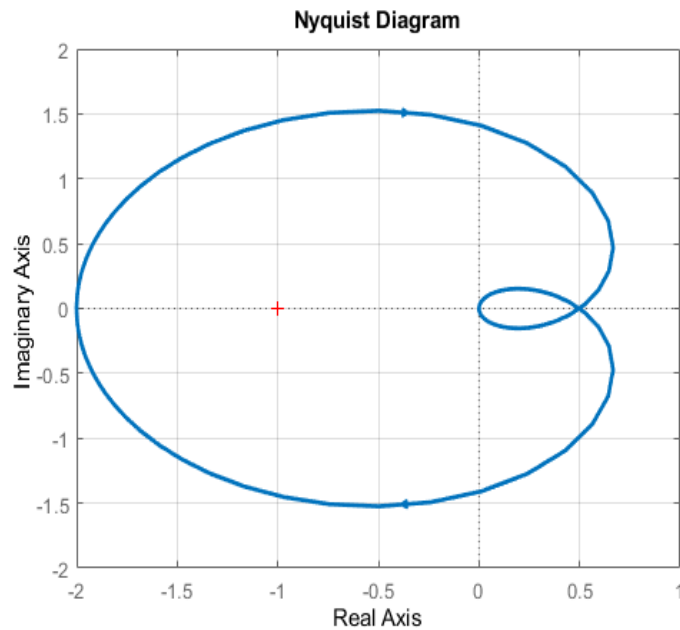
1. For unity feedback closed loop system, the forward path transfer function $G(s)$ is given as,

$$G(s) = K \frac{(s - 2)}{(s + 1)^2}$$

(A) Draw the Nyquist plot.

(B) Determine the condition for stability of the closed loop system using Nyquist plot.

INDICATIVE SOLUTION: Nyquist plot, assuming $K = 1$.



2. Consider a closed loop system having loop transfer function $G(s)H(s)$. Determine the range of dead time ' T ' such that the closed loop system remains stable. Given,

$$G(s)H(s) = \frac{e^{-sT}}{s(s+1)}$$
